Lexicographic ordering Ranking Preservation under Bounded Score Perturbations

Research Architecture Runtime
`operator/lexicographic-ordering`

2026-07-29

Abstract

`lexicographic-ordering` is certified by pairwise ranking preservation when all gaps exceed 2ε. Lean `LEAN_FULL` via `OrderStat.Ranking`.

This theorem is: `Reduction`.

1 Problem

The `lexicographic-ordering` operator depends on the pairwise order of scores $s \in \mathbb{R}^n$. The certified object is the full pairwise ranking (and thereby any ranking-determined selection such as top-k index sets).

2 Stability notion

Perturbations: $\|\delta\|_\infty \leq \varepsilon$. Stability: if $\mathrm{AllGapsExceed}(s, 2\varepsilon)$, then $s_i < s_j \Leftrightarrow (s+\delta)_i < (s+\delta)_j$ for all i, j. Sharpness: a gap $\leq 2\varepsilon$ admits a collision adversary.

3 Definitions

Definition 1 (All-gaps). *$|s_p - s_q| > g$ for $p \neq q$.*

Assumptions. $n \geq 2$, $\varepsilon \geq 0$, and (for invariance) all pairwise gaps $> 2\varepsilon$.

4 Theorem

Theorem 2 (Ranking invariance). *If $\mathrm{AllGapsExceed}(s, 2\varepsilon)$, then for every $\|\delta\|_\infty \leq \varepsilon$ and all i, j, $s_i < s_j \Leftrightarrow s_i + \delta_i < s_j + \delta_j$.*

Theorem 3 (Sharpness). *If some distinct pair has $|s_i - s_j| \leq 2\varepsilon$, an admissible δ forces a value collision.*

5 Intuition

Relative ℓ_∞ moves cost at most 2ε. Gaps larger than 2ε cannot reverse; smaller gaps can be closed by a midpoint push.

6 Examples

Example 4. *Scores* $(0, 3, 8)$, $\varepsilon = 1$*: all gaps* > 2*, so the ranking is stable. If scores are* $(0, 1.5)$ *and* $\varepsilon = 1$*, the gap* $1.5 \leq 2$ *admits a midpoint tie.*

7 Proof

Fix $\|\delta\|_\infty \leq \varepsilon$. If $i = j$ both sides of the claimed equivalence are false for strict $<$. If $i \neq j$ and $|s_i - s_j| > 2\varepsilon$, the estimate $(s_i + \delta_i) - (s_j + \delta_j) \geq (s_i - s_j) - 2\varepsilon$ (and its swap) yields $s_i < s_j \Leftrightarrow s_i + \delta_i < s_j + \delta_j$ (Theorem 2).

For sharpness, take $i \neq j$ with $|s_i - s_j| \leq 2\varepsilon$ and the midpoint $\delta_i = -(s_i - s_j)/2$, $\delta_j = (s_i - s_j)/2$ (elsewhere 0). Then $\|\delta\|_\infty \leq \varepsilon$ and $s_i + \delta_i = s_j + \delta_j$ (Theorem 3).

8 Formal statement

Lean module: `Research.Operators.LexicographicOrdering.Preservation`. Source file: `lean/Research/Opera`
Theorems: `lexicographic_ordering_margin_invariance`, `lexicographic_ordering_margin_sharpness`.
Reduction of `Research.Operators.OrderStat.Ranking`. Certificate: `lean/certificates/lexicographic-orde`
Status: `LEAN_FULL` on Mathlib $\mathbb{R}$ (`REAL_MATHLIB`).

9 Proof dependencies

- `Research.Operators.OrderStat.Ranking` / shared pairwise lemma from `KthMargin`.
- Midpoint collision adversary.
- No other operator theorems required.

10 Consequences

Ranking stability implies stability of any functional of the pairwise order used by `lexicographic-ordering`. Related operators: `top-k`, `sorting`, `rank`.